



Data Collection and Sampling

Megan Ayers
Stat 141 | Fall 2026
Friday, Week 3

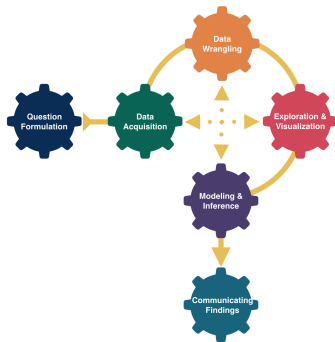
1

Goals for Today

- Quick clarification: independent vs disjoint
- Discuss principles of data collection/acquisition
- Investigate 3 methods of drawing random samples

2

Moving into Data Collection



3

Who are the data supposed to represent?

Every statistical investigation should clearly identify and compare:

- The **population** to be studied
- The **sample** from which measurements (data) will be taken

Key questions:

- What evidence is there that the sample is **representative** of the population?
- Who is present? Who is absent?
- Who is overrepresented? Who is underrepresented?

4

Who are the data supposed to represent?



In a **census**, we have data on the entire population!

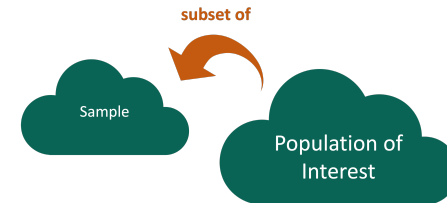
Lets us calculate summary statistics about the whole population (we call these **parameters**).

But usually, we don't have the money, time, or ability to do this :(

- The 2020 census cost about \$13.7B!!

5

Who are the data supposed to represent?

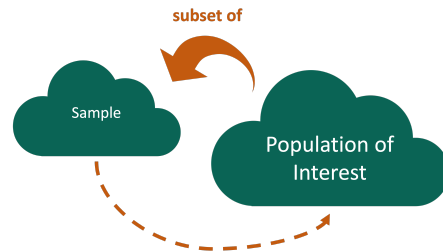


Instead, we use a **sample** of the population, and use the sample to draw conclusions about the **population**.

We calculate **statistics** based on our **sample** and use these as **estimates** of the population **parameters** that we really want to know.

6

Who are the data supposed to represent?

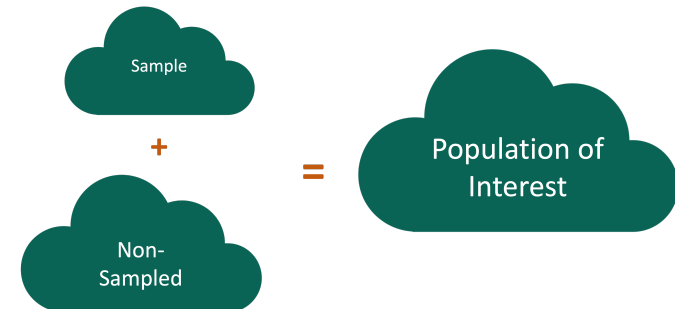


Key questions:

- What evidence is there that the **sample** is **representative** of the **population**?
- Who is present? Who is absent?
- Who is overrepresented? Who is underrepresented?

7

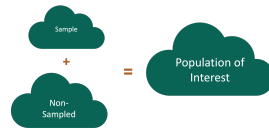
Sampling Bias



Sampling bias: When certain individuals are more likely to be sampled than others

8

Sampling Bias



Sampling bias: When certain individuals are more likely to be sampled than others

Q: Consider a telephone poll for an election - where might we get sampling bias?

- Non-response: individual can't or won't contribute
- Undercoverage: some groups are less likely to be called
- Inaccurate response
- Self-selection: membership in the sample is voluntary
- Convenience: selecting a convenient but non-representative block to sample

9

Sampling Bias Example

The **Literary Digest** was a political magazine that correctly predicted the presidential outcomes from 1916 to 1932. In 1936, they conducted the most extensive (to that date) public opinion poll. They mailed questionnaires to over 10 million people (about 1/3 of US households) whose names and addresses they obtained from telephone books and vehicle registration lists.

Population of Interest:

Sample:

Sampling bias:

02:30

10

Sampling Bias Example

We want to know how Portlanders feel about a new coffee shop in Woodstock.

- The coffee shop has a Yelp rating of 3.5/5 stars with 10 reviews.

Q1: Can we conclude that a typical Portlander would rate this coffee shop at 3.5 stars?

Q2: What sources of bias are present in this sample?

Q3: A year later, the coffee shop still has 3.5 stars, but 1000 reviews. Does the verdict change?

Q4: A second coffee shop opens up nearby with a Yelp rating of 4 stars and 1000 reviews. Can we conclude Portlanders prefer the second restaurant to the first?

03:00

11

Random Sampling

Use random sampling (a random mechanism for selecting cases from the population) to remove sampling bias.

Types of random sampling

We'll explore 3 types of random sampling

- Simple random sampling
- Cluster sampling
- Stratified random sampling

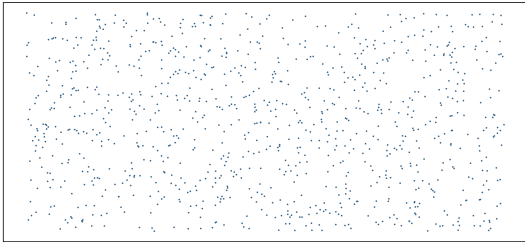
12

Simple Random Sampling

Motivating question: what is the average amount of student loan debt in Oregon?

Simple Random Sampling: Imagine that a unique ID for each student *in the population* is written on a slip of paper...

- Shuffle the slips of paper in a bowl
- Draw n IDs/slips one-by-one to create a sample



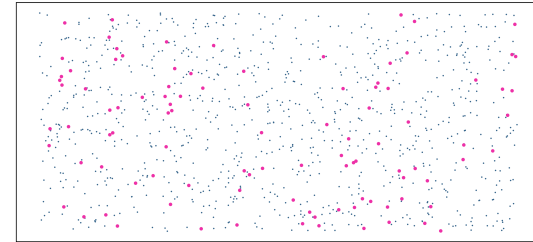
13

Simple Random Sampling

Motivating question: what is the average amount of student loan debt in Oregon?

Simple Random Sampling: Imagine that a unique ID for each student *in the population* is written on a slip of paper...

- Shuffle the slips of paper in a bowl
- Draw n IDs/slips one-by-one to create a sample



14

Simple Random Sampling

Consequences:

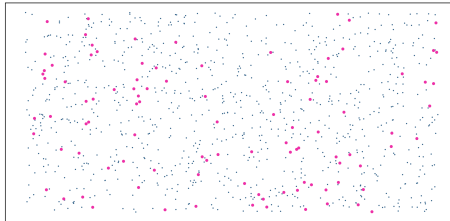
- Every member of the population has an **equal chance of being selected** for the sample
- There is **no inherent correlation** between any two members of the sample

Q: Can a simple random sample be non-representative?

A: Yes, even if all goes as planned!

- For large sample sizes, it's unlikely
- The sample will be **representative on average**

Q: Why aren't all samples generated using simple random sampling?



15

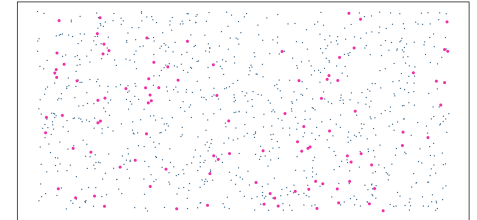
Simple Random Sampling

Advantages:

- Relatively simple to interpret and analyze
- Non-biased (in theory)

Disadvantages:

- May not be as "precise" as other sampling techniques
- Can be difficult to perform in practice



16

Stratified Random Sampling

Motivating question: what is the average amount of student loan debt in Oregon?

Stratified Random Sampling: “Strata” are made up of similar individuals, then simple random samples are taken from each stratum.

- e.g. define strata based on public vs private college and family income ranges



17

Stratified Random Sampling

Stratified Random Sampling: “Strata” are made up of similar individuals, then simple random samples are taken from each stratum.

Advantages:

- Can be more “precise” than simple random sampling, requiring lower sample size
- Hedges against non-representative samples

Disadvantages:

- Statistical analysis is more complex
- Strata creation isn’t always straightforward (need additional data, and to have compelling reasons for strata definitions)



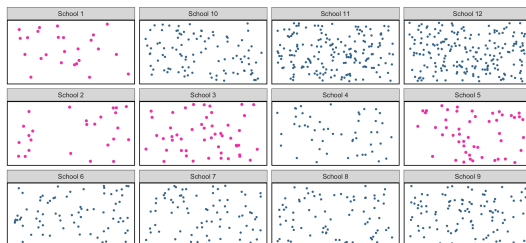
18

Cluster Random Sampling

Motivating question: what is the average amount of student loan debt in Oregon?

Cluster Random Sampling: “Clusters” are non-homogeneous. We take a simple random sample of the clusters, and use all observations in those clusters as the sample.

- e.g. we take a simple random sample of schools, include all students in those schools in the sample



19

Cluster Random Sampling

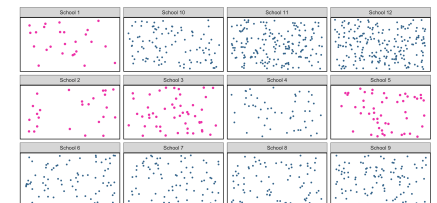
Cluster Random Sampling: “Clusters” are non-homogeneous. We take a simple random sample of the clusters, and use all (or some) observations in *those* clusters as the sample.

Advantages:

- Useful when it’s difficult/impossible to exhaustively list the population
- Often more time/cost effective per n
- Useful when population is naturally concentrated in heterogeneous groups

Disadvantages:

- Less precise than simple or stratified sampling
- Statistical analysis is more complicated
- Natural clusters may not always exist



20

National Health and Nutrition Examination Survey



Mission: "Assess the health and nutritional status of adults and children in the United States."

How are these data collected?

21

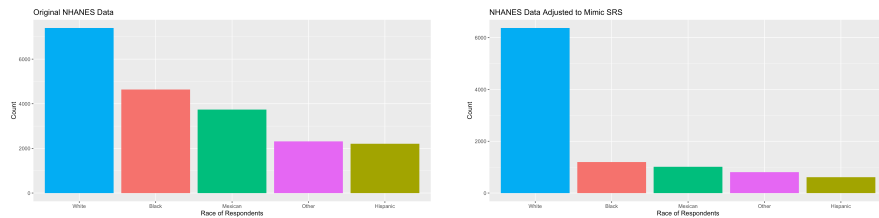
NHANES Sampling Design

- **Stage 1:** US is stratified by geography and distribution of minority populations. Counties are randomly selected within each stratum.
- **Stage 2:** From the sampled counties, city blocks are randomly selected. (City blocks are clusters.)
- **Stage 3:** From sampled city blocks, households are randomly selected. (Households are clusters.)
- **Stage 4:** From sampled households, people are randomly selected. For the sampled households, a mobile health vehicle goes to the house and medical professionals take the necessary measurements.

Why don't they use simple random sampling?

22

Careful Using Non-Simple Random Sample Data



23

Detour: Data Ethics

24

Data Ethics

“Good statistical practice is fundamentally based on transparent assumptions, reproducible results, and valid interpretations.” – Committee on Professional Ethics of the American Statistical Association (ASA)

The ASA has created “**Ethical Guidelines for Statistical Practice**”

- These guidelines are for EVERYONE doing statistical work.
- There are ethical decisions at all steps of the Data Analysis Process.
- We will periodically refer to specific guidelines throughout this class.

“Above all, professionalism in statistical practice presumes the goal of advancing knowledge while avoiding harm; using statistics in pursuit of unethical ends is inherently unethical.”

25

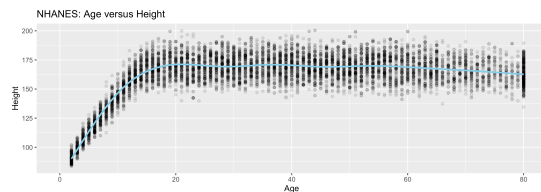
Responsibilities to Research Subjects

“The ethical statistician protects and respects the rights and interests of human and animal subjects at all stages of their involvement in a project. This includes respondents to the census or to surveys, those whose data are contained in administrative records, and subjects of physically or psychologically invasive research.”

26

Responsibilities to Research Subjects

Why do you think the **Age** variable maxes out at 80?



“Protects the privacy and confidentiality of research subjects and data concerning them, whether obtained from the subjects directly, other persons, or existing records.”

27

Recap

- We often want to learn about an entire **population** (ex. all US voters) but only have access to a **sample** (ex. 1000 survey respondents)
- We can use a sample to learn about the population **IF** it's **representative** of the population
- **Sampling bias** occurs when systematic differences between our sample and the population lead us to make incorrect conclusions
- Three ways to obtain a representative sample:
 - **Simple random sampling** (select cards from *one* hat)
 - **Cluster random sampling** (select *some* hats from a group of many *similar* hats, take their cards)
 - **Stratified random sampling** (select some cards from *every* hat in a group of very *different* hats)

Next time:

- Study design

28